Cristobalite margin, G(cristobalite) − min G(quartz, tridymite)
Margin (meV/atom)
2.0
1.5
1.0
0.5
0.0
−0.5
DPA: 1824 K, after a tridymite interval
DPA + volume (equal-block): never stable
reference 1543 K
below 0: cristobalite is the stable phase
Three source phases
DFT reference (PBE-D3(BJ))
DPA
DPA + volume (equal-block)
ΔG MAE (meV/atom)
Stable-phase accuracy
Transitions hit / miss / false
reference 1543 K
DFT reference (PBE-D3(BJ))
Two source phases · β-quartz, P6₃/mmc tridymite
Volume only 1583 K 0.276 41.96% 0 / 1 / 1
DPA 1583 K 0.853 97.57% 1 / 0 / 0
DPA + volume (per-dimension) 1583 K 0.845 97.57% 1 / 0 / 0
DPA + volume (equal-block) 1583 K 0.401 97.57% 1 / 0 / 0
Three source phases · + C222₁ tridymite
Volume only 2016 K 1.435 41.96% 0 / 1 / 1
DPA 1583 K 1824 K 0.315 82.96% 0 / 1 / 2
DPA + volume (per-dimension) 1583 K 1832 K 0.315 82.41% 0 / 1 / 2
DPA + volume (equal-block) 1583 K 0.408 41.96% 0 / 1 / 1
1000
1200
1400
1600
1800
2000
2200
2400
Temperature (K)
β-quartz
β-tridymite (P6₃/mmc)
β-cristobalite
wrong stable phase
reference transition (1543 K)
DPA, two vs three sources